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PI Name:	Valles, James M. Ph.D.		
Project Title:	Magnetic Field Gradient Levitation System For Physics and Biophysics		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Biological physics		
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth's gravity		

We are developing a Magnetic Field Gradient Levitation (MFGL) apparatus as a ground based system for simulating a low or variable gravity environment for samples made of diamagnetic materials. The system design takes advantage of recent technological advances that have made it possible to create a superconducting solenoid with a room temperature bore that generates a magnetic force strong enough to levitate or cancel the effects of gravity in common organic materials (e.g. water, proteins, polypropylene, etc.). The apparatus will be conducive for pre-space flight studies of the sources of gravitational sensitivity in biological systems and for the investigation of soft condensed matter systems in the virtual absence of gravity. Initial experimental tests of the apparatus will be performed on two biological systems, frog embryos of the species, Xenopus laevis and paramecium. These systems have shown well-defined gravitational sensitivity in experiments on the Space Shuttle.

Task Description:	The technique of Magnetic Field Gradient Levitation (MFGL) is being applied successfully in low temperature studies of fundamental physics in pure superfluid helium and in tests of potential pure superfluid helium experiments for space flight. The primary objective of this project is to extend the MFGL technique to studies of organic physical and biological systems that are composed of more than one type of material and exist in a room temperature environment. We will work to determine the efficacy of MFGL as a means of simulating low gravity on the ground for biological systems and as a means of screening potential systems for experiments in space. In addition, we will endeavor to develop insight into the potential of methods for manipulating systems in the ultra low gravity environment of space using magnetic forces. We take our ability to know up from down for granted relying on the force of gravity to provide the cues. How we, as biological systems, sense and convert these cues into a biological response is important to understand. For example, it is of practical interest to know how prolonged exposure to the low gravity environment of space leads to the documented adverse effects on the immune and skeletal systems of astronauts. In principle, careful biology experiments during spaceflight can reveal the answers. Knowing which experiments to do, however, often requires a great deal of spade work that is impractical to carry out in space. Consequently, experimental methods for simulating low gravity conditions on earth are very desirable. This work will determine the efficacy of Magnetic Field Gradient Levitation (MFGL) as such a method. It is also anticipated that MFGL will be useful for screening other potential space shuttle experiments in physics and biology and may lead to the development of novel magnetic methods of manipulating objects in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We developed a non-invasive method for exerting weak forces on biological cells that has potential application in investigations of how cells transduce stresses. The method employs the same magnetic techniques as those we use to simulate different strength gravity environments for biological systems.
Task Progress:	Our work has reached the goal of establishing a magnetic field gradient levitation system as a means to simulate variable gravity environments, including zero gravity, for biological systems. We have shown that the swimming protist, paramecium, responds to magnetically simulated gravity in the same manner that it responds to earth's gravity and centrifugation. This advance paves the way for earthbound investigations of life in low or zero gravity. Moreover, we demonstrated that the intense magnetic fields involved can also be employed to align and direct the motion of swimming microorganisms without deleterious effects. This capability allows new methods for collecting micro-organisms and controlling their behavior for experiments. Through the course of this work, apparatus for two different magnet systems were developed and employed. A superconducting solenoid with a room temperature experimental space was acquired from American Magnetics Incorporated and installed in the Valles lab. It provides magnetic forces sufficiently strong to balance the weight of water and many proteins and thus, is suitable for our variable gravity studies. An optical system for viewing the biological specimens in the confined experimental space (a cylinder with a diameter of 11 mm) was designed using borescope technology. It has sufficient resolution for tracking swimming paramecia and euglena. The second magnet system is the resistive magnet in Cell 5 at the National High Magnetic Field Laboratory in Tallahassee, Florida. This system produces magnetic forces that are 3 1/2 times stronger than thos e attainable with the superconducting solenoid and boasts a significantly larger (50 mm diameter) experimental space. An apparatus for holding and viewing samples while maintaining them at a constant temperature was developed and employed for a series of experiments on swimming paramecia in simulated gravity. Using the NHMFL magnet and adjusting the solutions in which the paramecia swim, we achieved effective gravity conditions ranging from 10 times earth's gravity to inverted gravity up to 8 times earth's gravity including effectively zero gravity. This research on paramecium in simulated gravity served as the primary topic of the PhD. dissertation of Dr. Karine Guevorkian. Karine received her degree from Brown University this past spring (May 2006). Through her research, she received training in experimental techniques, computer modelling, computer analysis of data, biological physics and how to communicate scientific results. Karine will begin postdoctoral research in biophysics at Institut Madame Curie shortly. A second student and PhD candidate, Hung Nyugen has embarked on comparative studies of another swimming microorganism, Euglena. In addition, a few undergraduate science majors at Brown have performed small related projects to give them scientific training. The research results have been the subject of a number of invited talks at National and International meetings given by the PI, Valles. In addition, the results appear in a number of publications. The most notable publication is about to appear in the prestigious journal Proceedings of the National Academy of Science. It describes the experiments in which we observed paramecia's reaction to enhanced, reduced and inverted gravity. These experiments showed that Paramecium exhibits a linear response to magnetically simulated hyper-gravity up to 5g, modifying its swimming as it would in the hyper-gravity of a centrifuge. Moreover, experiments from 0 to -5 g show that the response is symmetric, implying that the regulation of the swimming speed is primarily related to the buoyancy of the cell. The response becomes non-linear for simulated buoyancy higher than 5 g. At 10 g, many paramecia “stall” (i.e. swim in place against the force) exerting a maximum propulsion force estimated to be 0.7 nN. These findings establish a general technique for applying continuously variable forces to cells or cell populations suitable for exploring their force transduction mechanisms. The second major publication appeared in Biophysical Journal and it related how paramecia align their swimming with intense magnetic fields. We related how the trajectories of Paramecium caudatum align with intense static magnetic fields greater than 3 Tesla. Otherwise straight trajectories, curve in magnetic fields and eventually orient parallel or anti-parallel to the applied field direction. Neutrally buoyant immobilized paramecia also align with their long axis in the direction of the field. We presented a model of this magneto-orientation as a strictly passive, non physiological, response to a magnetic torque exerted on the diamagnetically anisotropic components of the paramecia. We suggested that magnetic fields can be exploited as a novel, non-invasive, quantitative means to manipulate swimming populations of unicellular organisms. In a third publication, we described our apparatus for performing experiments on swimming microorganisms at the National High Magnetic Field Laboratory with the hope that others could benefit from our designs. The development of an optical system that was non-magnetic and functioned with high resolution and a large field of view in the confines of the 50 mm space was the main news. A fourth publication "Manipulating Cells with Static Magnetic Fields? connected

	our work to the community of researchers interested in using magnetic fields to manipulate materials. Through this project we developed and established a novel method for simulating variable gravity on earth which is suitable for experiments on the effect of weightlessness on cellular systems. More generally, the research enhanced our nation's infrastructure by providing scientific research training and exposure to new technologies to both undergraduate and graduate level scientists.
Bibliography Type:	Description: (Last Updated: 04/16/2007)
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